

Unit 5

SYSTEM SOFTWARE

38 items · 245 marks

Syllabus sections in this unit

5.1 Operating Systems

5.2 Language Translators

Unit 5: System Software

1. M/J 26/P11/Q2/a,b,c

- 2 A computer programmer uses language translators. Two types of translator are an interpreter and a compiler.

(a) Explain why language translators are needed when programming.

[2]

(b) Describe **two** benefits of using an interpreter instead of a compiler when developing a program.

1

2

[4]

(c) Identify **one** other type of translator.

[1]

2. M/J 26/P12/Q7/a,b

7 A student is writing a computer program for a school project.

(a) The student uses a computer that comes with an Operating System (OS) and utility software.

(i) Identify **two** tasks the OS performs as part of process management.

1

2

[2]

(ii) Identify **two** tasks the OS performs as part of memory management.

1

2

[2]

(iii) Explain the reasons why defragmentation software improves the performance of a computer.

[3]

(b) The student is using an Integrated Development Environment (IDE).

(i) Explain how the student can use debugging tools in the IDE.

[4]

(ii) The high-level language the student is using can be partially compiled and partially interpreted.

Explain the reasons why some high-level languages are partially compiled and partially interpreted.

[2]

3. M/J 26/P13/Q5/a,b,c

- 5** A student uses a laptop for school work. The laptop contains system software including an Operating System (OS) and utility software.

(a) Hardware management is one key management task that the OS performs.

- (i)** Identify **two** tasks the OS performs as part of hardware management.

1

2

[2]

- (ii)** Identify **two** other key management tasks that the OS performs.

1

2

[2]

(b) A disk formatter is an example of utility software.

- (i)** State the purpose of utility software in a computer.

[1]

- (ii)** Describe the purpose of a disk formatter.

[2]

(c) The student is writing a computer program in a high-level language.

The student imports routines from program libraries into their program.

- (i)** Define the term program library.

[1]

- (ii)** Describe **one** benefit of the student using a Dynamic Link Library (DLL) instead of a non-Dynamic Link Library.

[2]

4. M/J 26/P21/Q2/b

also Unit 12 (1 mark); this unit 1 mark

- 2** A washing machine contains:

- two trays for the washing powder
- a user interface display
- five sensors.

The washing machine control program includes the following variables:

variable	description
MainsOn	TRUE when the mains power is on
WaterOn	TRUE when the water supply to the machine is turned on
DisplayOn	TRUE when the user interface display is active
Tray1	TRUE when it detects washing powder in tray 1
Tray2	TRUE when it detects washing powder in tray 2

- (b) The user will select from three possible wash options.

The pseudocode for the washing machine program contains a procedure `WashCycle()`

```
PROCEDURE WashCycle()
    // declaration of variables has been omitted
    // TimeSoFar is a global variable which changes
    // as the procedure is executed
    Finished ← FALSE
    TimeSoFar ← 0.0
    OUTPUT "Option (1/2/3) "
    INPUT OptionNumber
    CASE OF OptionNumber:
        // WashTime is the total wash time in minutes
        1 : WashTime ← 60
        2 : WashTime ← 90
        3 : WashTime ← 120
    ENDCASE
    WHILE Finished = FALSE
        TimeRemaining ← WashTime - INT(TimeSoFar)
        
    ENDWHILE
ENDPROCEDURE
```

- (i) The programmer uses an Integrated Development Environment (IDE) to develop and run the washing machine program.

State the IDE feature to display the current value for any variable during the execution of the program.

[1]

- (ii) When option 3 is selected, the programmer interrupts execution when the `TimeSoFar` is 17.6 minutes.

State the current value for `TimeRemaining`

[1]

5. M/J 26/P23/Q2/d

also Unit 12 (1 mark); this unit 1 mark

- 2 A control program is being developed for a robot floor vacuum cleaner. The settings for the robot can be changed and include:

- the strength of the suction (numerical values: 0, 1, 2, 3, 4 or 5)
- the total vacuum cleaning time (a whole number of minutes).

The robot has a number of sensors.

The control program includes the variables:

variable	description
MainsOn	TRUE when the mains power is on
DisplayOn	TRUE when the user display is shown
Suction	TRUE when the program detects that the user has selected a non-zero value for the strength of the suction

(d) The pseudocode for the control program contains the procedure `CleanProgram()`

```

PROCEDURE CleanProgram()
// declaration of variables has been omitted

    Finished ← FALSE
    OUTPUT "Total clean time (minutes) "
    INPUT  TotalCleanTime
    OUTPUT "Suction strength (0,1,2,3,4,5) "
    INPUT  SuctionStrength

    WHILE Finished = FALSE
        // TimeSoFar is a global variable which changes
        // as the procedure is executed
        // TimeSoFar is the time the robot has been cleaning (in seconds)

        TimeRemaining ← TotalCleanTime - INT(TimeSoFar/60)

        }
    ENDWHILE

    }
ENDPROCEDURE

```

- (i) When the program is under development and running, the programmer can temporarily stop the execution at a set point in the program.

Identify this feature of the Integrated Development Environment (IDE).

[1]

- (ii) The programmer runs the program.

- The user is prompted and inputs the value 40 (total clean time in minutes).
- The user is prompted and inputs the value 3 (for the suction strength).
- The programmer interrupts its execution; the `TimeSoFar` is shown in a report window as 125 (the cleaning time so far in seconds).

State the current value of `TimeRemaining`

[1]

6. O/N 25/P11/Q5/a,b,c

5 The Operating System (OS) is responsible for hardware and security management in a computer system.

(a) (i) State **two** tasks that are performed by hardware management.

1

2

[2]

(ii) State **two** tasks that are performed by security management.

1

2

[2]

(b) (i) Describe **one** benefit of using library routines when developing software.

[2]

(ii) Describe **one** drawback of using library routines when developing software.

[2]

(c) Identify **two** presentation features found in a typical Integrated Development Environment (IDE).

1

2

[2]

7. O/N 25/P12/Q8/b(ii)

8 A restaurant accesses software and stores data using cloud computing.

(b) The restaurant uses touchscreens as input devices to allow staff to select items from a menu.

(ii) The image files that are displayed on the touchscreen have been corrupted.

Identify **two** examples of utility software that could be used to recover the images.

1

2

[2]

8. O/N 25/P12/Q10/a

10 A programmer is developing a computer program.

(a) Explain how the programmer can first use an interpreter **and** then a compiler to develop the computer program.

Interpreter

Compiler

[4]

9. O/N 25/P13/Q8/a,b,d

- 8** Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

(a) State **two** disadvantages of using a compiler compared to an interpreter during program development.

1

2

[2]

(b) Explain how a programmer benefits from using program libraries during program development.

[3]

(d) Memory management is one key management task performed by an OS.

Give **three** other key management tasks that are performed by an OS.

1

2

3

[3]

10. O/N 25/P21/Q1/c

- 1** A program is being developed to meet a particular customer requirement.

(c) During the alpha testing stage, an Integrated Development Environment (IDE) is used to help locate an error that has been identified. The IDE report window feature is used to examine the values assigned to variables.

Explain how **two** other IDE features are used together with the report window feature to help locate the error.

[3]

11. M/J 25/P11/Q6/b

- 6** A programmer is buying a new computer.

(b) Both computers have an Operating System (OS).

Describe the purpose of an OS in a computer.

[5]

12. M/J 25/P13/Q3/a

3 A programmer is writing a program in a high-level language.

(a) Complete the description of compilers and interpreters by writing the missing words.

A compiler checks all the code before attempting to translate the program. If any errors are found, they are all reported at the same time and the program does not translate or run. If there are no errors found, the compiler produces which can run without access to the

An interpreter translates one line of code and then runs it, before moving to the next line of code. If the line of code has an error, the interpreter and displays the error. The programmer can correct the error and then the interpreter continues translating from that point.

[4]

13. O/N 24/P11/Q4/a,c,d

4 A student uses a laptop to write a program that is saved as a text file.

(a) The laptop has utility software and an Operating System (OS).

(i) Describe the file management tasks carried out by an OS.

[2]

(ii) Explain the need for back-up software.

[2]

(c) The student used a program library when writing their program.

Explain the benefits to the student of using library files when writing a program.

[3]

(d) The program code is written using an Integrated Development Environment (IDE).

(i) One presentation feature found in a typical IDE is prettyprint.

Identify **and** describe **one other** presentation feature found in a typical IDE.

Feature

Description

[2]

(ii) One debugging feature found in a typical IDE is single stepping.

Identify **and** describe **one other** debugging feature found in a typical IDE.

Feature

Description

[2]

14. O/N 24/P12/Q3/c

3 A student has a computer.

- (c) The computer has an Operating System (OS). One of the key management tasks of the OS is process management.

Describe the process management tasks performed by an OS.

[4]

15. O/N 24/P12/Q4/a,b

4 A program is written in a high-level language by a team of three programmers using an Integrated Development Environment (IDE).

- (a) Describe how the programmers can use the debugging features of a typical IDE during the development of the program.

[4]

- (b) The programmers created a new program library whilst developing the program.

Describe the benefits to the programmers of creating a program library.

[3]

16. O/N 24/P22/Q1/b,c

1 A program has been developed and released for general use. After a few months of use an error is detected where under certain circumstances it outputs an unexpected value.

- (b) The program contains a function `Lookup()`. After investigation, it is found that this is the function that sometimes returns an incorrect value.

An Integrated Development Environment (IDE) is used to help locate the error.

The IDE features of watch window, single stepping and breakpoint will be used.

Explain these features including the order that they will be used in to locate the error in `Lookup()`.

[3]

- (c) To solve the error a programmer decides to create a new module.

The design of the new module has been completed and the module is being coded.

Identify **two** features of an IDE that will help during the coding of this new module.

1

2

[2]

17. M/J 24/P11/Q3/a,b

3 A software developer is writing a computer program.

- (a) The developer uses an interpreter while writing the program code because it is easier for debugging.

Explain **one** reason why it is easier to debug the program code using an interpreter instead of a compiler.

[2]

- (b) The program is ready to be sold to customers.

The developer uses a compiler because it creates an executable file.

Explain the reasons why the need to create an executable file makes the compiler the appropriate choice when the program is complete.

[3]

18. M/J 24/P12/Q6

6 A computer has an Operating System (OS).

Memory management and process management are two OS tasks.

Explain how memory management **and** process management support multi-tasking.

[4]

19. M/J 24/P12/Q8/a,b,c

8 A programmer uses an Integrated Development Environment (IDE) to write a computer program. The IDE has both a compiler and an interpreter as built-in translators.

- (a) The programmer decides to use the compiler when testing the final program.

Describe the benefits of using the compiler during testing.

[2]

- (b) IDEs have many features other than built-in translators.

Complete the table by identifying **one other** common IDE feature that can be used for each purpose. Describe how each feature helps the user during program development.

Each feature must be different. Do **not** give translator as one of your features.

Purpose	IDE feature	Description
for coding		
for presentation		
for debugging		

[6]

(c) The programmer uses program libraries when developing the program.

Describe **two** benefits to the programmer of using program libraries.

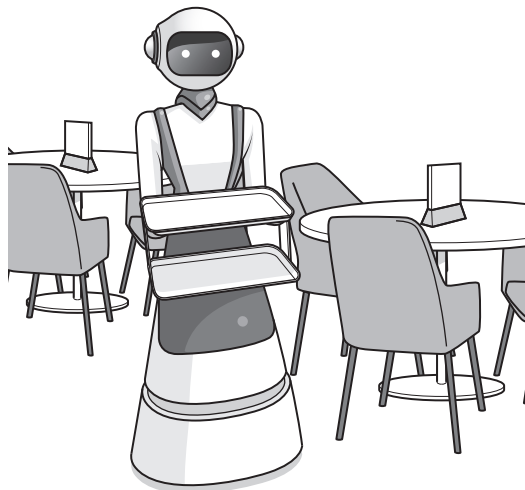
1

2

[2]

20. M/J 24/P13/Q7/e

7 Robots are used to serve food and drink to customers at a restaurant.



(e) Program libraries were used when writing the robot's software.

(i) State what is meant by a **program library**.

[1]

(ii) Some program libraries include Dynamic Link Library (DLL) files.

Describe the benefits of a programmer using a library with DLL files instead of using a library that does not include DLL files.

[4]



21. O/N 23/P11/Q6/a,b,c

- 6** A programmer uses both a compiler and an interpreter to translate a program written in a high-level language.

(a) Describe the advantages of using the interpreter compared to the compiler to translate the program.

[4]

(b) State **one** reason why some high-level languages are partially compiled and partially interpreted.

[1]

(c) (i) Identify **two** features that support the visual presentation of the code in a typical Integrated Development Environment (IDE).

1

2

[2]

(ii) Identify **two** features that support the debugging of the code in a typical IDE.

1

2

[2]

22. O/N 23/P12/Q8/a,b

- 8** A computer has an Operating System (OS).

(a) State **one** purpose of the Operating System.

[1]

(b) The Operating System has utility software.

The purpose of some utility software is to improve security.

Identify **one** example of utility software that is **not** intended to improve security.

Explain why this software is needed.

Utility software

Explanation

[3]

23. O/N 23/P13/Q5/a,b(ii)

- 5 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

(a) Describe the following features of a typical IDE.

Context-sensitive prompts

Single stepping

[4]

(b) The program is distributed by downloading the source code and its library files from a web server.

(ii) The program files are stored on a new hard disk after they have been downloaded.

Describe the reasons why a hard disk formatter is needed for the new hard disk.

[3]

24. M/J 23/P11/Q3/a,b,c

- 3 A computer has an Operating System (OS).

(a) Describe how the Operating System manages the peripheral hardware devices of the computer.

[4]

(b) Hardware management is one key management task carried out by the Operating System.

Identify **two other** key management tasks carried out by the Operating System.

1

2

[2]

(c) The Operating System has utility software including defragmentation software.

Explain how defragmentation can improve the performance of the computer.

[3]

25. M/J 23/P12/Q5/d

- 5 A student has purchased a new laptop.

(d) The laptop has systems software.

(i) Describe how the Operating System (OS) manages processes in the computer.

[5]

(ii) Describe the purpose of utility software in a computer.

[2]

26. M/J 23/P12/Q7/a,b,c

7 A software developer is working in a team writing a program for a client.

(a) The developer is writing a new program library to be used by the other team members.

(i) Define the term **program library**.

[2]

(ii) Explain **two** benefits to the developer of choosing to create a Dynamic Link Library (DLL).

1

2

[4]

(b) The development team needs to use a translator whilst writing the program for the client.

Identify whether an interpreter or a compiler would be more appropriate at this stage of the program development.

Justify your choice.

Translator

Justification

[3]

(c) The development team uses an Integrated Development Environment (IDE).

Complete the table by describing the typical features found in an IDE.

Feature	Description
Breakpoints	
Dynamic syntax checks	
Context-sensitive prompts	
Single stepping	

[4]

27. M/J 23/P13/Q5/a

5 A programmer is developing a computer game in a high-level language to sell to the public.

(a) The programmer uses both an interpreter and a compiler at different stages of the development of the program.

(i) Explain the reasons why the programmer uses an interpreter while writing the program code.

[2]

(ii) Explain the reasons why the programmer uses a compiler when the program has been written.

[3]

28. M/J 23/P21/Q1/a

- 1 A programmer has written a program which includes the function `Calculate()`. When the program is run, the function returns an unexpected value.
- (a) Describe how a typical Integrated Development Environment (IDE) could be used to help debug the program to find the errors in the function `Calculate()`.

[4]

29. M/J 23/P22/Q5/c

- 5 A programmer has produced the following pseudocode to output the square root of the numbers from 1 to 10.

Line numbers are for reference only.

```
10 DECLARE Num : REAL
11 Num ← 1.0
...
40 REPEAT
41     CALL DisplaySqrt(Num)
42     Num ← Num + 1.0
43 UNTIL Num > 10
...
50 PROCEDURE DisplaySqrt(BYREF ThisNum : REAL)
51     OUTPUT ThisNum
52     ThisNum ← Sqrt(ThisNum)           // Sqrt returns the square root
53     OUTPUT " has a square root of ", ThisNum
54 ENDPROCEDURE
```

The pseudocode is correctly converted into program code.

Function `Sqrt()` is a library function and contains no errors.

The program code compiles without errors, but the program gives unexpected results. These are caused by a design error.

- (c) Describe how a typical Integrated Development Environment (IDE) could be used to identify this error.

[3]

30. O/N 22/P11/Q7/a,b

7 (a) State **two** benefits to a programmer of using Dynamic Link Library (DLL) files.

1

2

[2]

(b) Memory management is one of the tasks performed by an Operating System (OS).

Describe the ways in which memory management organises and allocates Random Access Memory (RAM).

[2]

31. O/N 22/P12/Q1/a,b

- 1 (a) Draw **one** line from each utility software to its most appropriate purpose.

Utility software	Purpose
virus checker	to reorganise files so they are contiguous
disk formatter	to scan for malicious program code
backup	to decrease the file size
disk repair	to initialise a disk
defragmentation	to create copies of files in case the original is lost
	to check for and fix inconsistencies on a disk

[5]

- (b) Compilers and interpreters translate programs written in a high-level language into a low-level language.

- (i) State **two** drawbacks of using a compiler compared to an interpreter during program development.

1

2

[2]

- (ii) Explain why high-level language programs might be partially compiled and partially interpreted.

[2]

32. O/N 22/P13/Q3

- 3 Draw **one** line from each Operating System (OS) management task to its most appropriate description.

OS Management task	Description
hardware management	dynamically allocates memory to processes
security management	marks unallocated file storage for availability
memory management	installs programs for devices connected to external ports
process management	validates user and process authenticity
	allows processes to transfer data to and from each other

[4]

33. O/N 21/P11/Q4/b(ii),d

4 Francis is starting his first job as a software developer for a multinational company.

(b) Francis is shown the software he will be working on. He is unfamiliar with the Integrated Development Environment (IDE) he is required to use.

(ii) A typical IDE provides debugging tools to support the testing of a program.

Identify **three** other tools or features found in a typical IDE to support the writing of the program.

1

2

3

[3]

(d) Francis's team use language translators.

Complete the descriptions of language translators by writing the missing words.

..... are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program.

They produce files that can be run without the source code.

..... translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line.

Assemblers are used to translate assembly code into

[4]

34. O/N 21/P12/Q7/b,c

7 A computer has system software.

(b) Describe the file management tasks that an Operating System performs.

[4]

(c) Identify **two** utility programs that can be used to improve the performance of a computer **and** state how they improve the performance.

1

2

[4]

35. O/N 21/P13/Q4/b(ii),d

4 Francis is starting his first job as a software developer for a multinational company.

(b) Francis is shown the software he will be working on. He is unfamiliar with the Integrated Development Environment (IDE) he is required to use.

(ii) A typical IDE provides debugging tools to support the testing of a program.

Identify **three** other tools or features found in a typical IDE to support the writing of the program.

1

2

3

[3]

(d) Francis's team use language translators.

Complete the descriptions of language translators by writing the missing words.

..... are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program.

They produce files that can be run without the source code.

..... translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line.

Assemblers are used to translate assembly code into

[4]

36. M/J 21/P11/Q2/a,b

2 Bingwen's computer comes with an Operating System and utility software.

(a) Draw **one** line from each utility software to its correct description.

Utility software	Description
	Scans software for errors and repairs the problems
Disk formatter	Moves parts of files so that each file is contiguous in memory
Defragmentation	Creates a copy of data that is no longer required
Back-up	Sets up a disk so it is ready to store files
Disk repair	Scans for errors in a disk and corrects them
	Creates a copy of data in case the original is lost

[4]

(b) Identify **four** key management tasks that the Operating System will perform.

1

2

3

4

[4]

37. M/J 21/P12/Q7/a,b

7 Jennifer is writing a computer program for her A Level homework.

- (a)** Jennifer uses a program library to help her write her computer program.

Describe how a program library can be used while writing a computer program.

[2]

- (b)** Jennifer uses an Integrated Development Environment (IDE) to write her computer program.

- (i)** The IDE allows Jennifer to use both an interpreter and a compiler while creating her computer program.

Describe the ways in which Jennifer can use **both** a compiler **and** an interpreter while developing the program.

[4]

- (ii)** Identify **two** debugging tools that a typical IDE can provide.

1

2

[2]

38. M/J 21/P13/Q2/a,b

2 Bingwen's computer comes with an Operating System and utility software.

(a) Draw **one** line from each utility software to its correct description.

Utility software	Description
	Scans software for errors and repairs the problems
Disk formatter	Moves parts of files so that each file is contiguous in memory
Defragmentation	Creates a copy of data that is no longer required
Back-up	Sets up a disk so it is ready to store files
Disk repair	Scans for errors in a disk and corrects them
	Creates a copy of data in case the original is lost

[4]

(b) Identify **four** key management tasks that the Operating System will perform.

1

2

3

4

[4]

Answers Section

1. M/J 26/P11/Q2/a,b,c

2(a)	1 mark each to max 2 - To convert a high-level language to a low-level language/assembly/machine code - Computers use machine code and programmers use high-level languages - To allow the computer to process the coded statements	2	Accept human language for HLL in MP2 MP3: Read is NE
2(b)	1 mark for each benefit and 1 mark for an appropriate expansion to max 4 (2 × 2) For example: - Amend the code or debug in real time - and see the changes immediately instead of having to recompile - Stops as soon as an error is encountered - so the position of the error can be easily seen - Only one error is displayed at a time // no dependent errors generated - individual errors are easier to identify/debug/fix - Incomplete code can be tested/run - so the program does not need to be complete to see if it works/see the output	4	
2(c)	1 mark for a correct answer Assembler	1	Correct answer only

2. M/J 26/P12/Q7/a,b

7(a)(i)	1 mark for each correct point to max 2 - Efficient allocation of resources to processes - Moving processes between states - Allocating time slices to processes - Scheduling processes // An example of a scheduling algorithm - Resolution of conflicts when two processes require the same resource - Allows communication between processes	2	Do not accept allows multi-programming – that is a benefit not a task resource management is NE Accept checking priorities as scheduling Take care that answers do not refer to tasks done by the kernel or by memory management.
7(a)(ii)	1 mark for each correct point to max 2 - Dynamically allocates RAM to processes - Frees up memory when the process closes - Makes sure two processes do not access the same memory - Accesses data from RAM when needed	2	Accept software or applications for processes in first mark point Take care that answers do not refer to tasks done by process management or by file management
7(a)(iii)	1 mark for each correct point to max 3 - With continued use files can be split and stored in different locations on the disk so time is spent by the disk head moving between locations to access all parts of file - Defragmentation software moves split files into contiguous blocks // defragmentation software collects all the free space into a contiguous block ... which means that the disk head does not need to move as much to access the next part of the file // ... which means that the disk head does not need to move as much when writing a file - After defragmentation data access/read/write takes less time, so performance is improved	3	Note: The question asks how defragmentation improves computer performance. So just a description of the process is NE

7(b)(i)	1 mark for each correct point to max 4 , mark in pairs (Single) Stepping ...the student runs one line of code at a time and looks at the output/results/flow before moving to run the next line - Breakpoints ...These are inserted in the code to stop the code at a specific point so that the student can check the flow/values / test sub-sections of the code (Variable) Watch window ...the student can view how/when the variables change during execution	4 Take care that the answer says how the student uses the tools and doesn't just describe each tool Award a maximum of two marks for names of debugging tools with no further suitable explanation Just check for errors is not enough for an explanation, that is what debugging means
7(b)(ii)	1 mark for each correct point to max 2 - Partially compiled programs can be used on different platforms as they are interpreted when run - The code is optimised for the CPU as machine code is generated at run time - It is more efficient to run as the source code does not need recompiling	2 Note: Candidate responses need to answer the question why?

3. M/J 26/P13/Q5/a,b,c

5(a)(i)	1 mark each to max 2 e.g. - Transfer data between peripheral devices - Translate data from I/O/peripheral devices - Loading and installation of device drivers - Hardware abstraction — hiding hardware differences from applications - Power management — sleep states, CPU throttling, battery control	2	
5(a)(ii)	1 mark each to max 2 e.g. - Memory management - Process management - File management - Security management // managing user accounts - Provide platform for running applications	2	
5(b)(i)	1 mark for: Perform housekeeping/maintenance tasks // improving/optimising overall performance	1	
5(b)(ii)	1 mark each to max 2 e.g. - Prepared a disk for future use by // initialising ...changing/setting file system // by example // directory structure // allocation tables ... naming the disk (volume label) ... creating a boot sector ...erasing all data ...creating partitions	2	
5(c)(i)	1 mark for: Set of pre-written / pre-compiled / pre-tested routines	1	

5(c)(ii)	1 mark for benefit 1 for expansion e.g. - Maintenance not needed to be done by the programmer ... because the DLL is separate from program // maintained by library creator - The calling program does not need recompilation by the programmer when a DLL file changes ... because the DLL file can be updated independently of the calling program ... updates will apply to all programs that use the DLL file (Executable) file is smaller ... because it does not contain (all) the library routines // code only linked at execution (Main) memory requirements for program is reduced ... because the DLL files are only loaded into memory when required ... one DLL file can be used by several running programs	2	
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4. M/J 26/P21/Q2/b

2(b)(i)	Report window // Watch window	1
2(b)(ii)	103	1

5. M/J 26/P23/Q2/d

2(d)(i)	<u>breakpoint</u>	1	
2(d)(ii)	38 (minutes)	1	

6. O/N 25/P11/Q5/a,b,c

5(a)(i)	1 mark per bullet point, max 2 marks - Installs <u>driver</u> software for devices connected to external ports - Manages communication between devices - Manages <u>hardware</u> interrupts	2
5(a)(ii)	1 mark per bullet point, max 2 marks - Prevents unauthorised access ... by providing authentication // by validating users and processes - Implements access rights and permissions - Makes provision for recovery of lost data - Carries out Operating System security updates as available - Carries out auditing and keeps logs of activity	2

5(b)(i)	One mark for benefit, one mark for appropriate matching expansion e.g. • Less code needs to be developed as code is pre-written • ... thereby reducing the time to write a program • The library routine will be pre-tested and documented • ... reducing the time taken to test a program // giving confidence that the routine works as it should • The code for the library routine can be written in a different programming language • ... which allows the use of special features of that language • The user's source code is more readable • ... as the library routine is simply called in the source code rather than being written out in full • The library could contain the code for complex routines • ... thus allowing the programmer to include modules they might not be able to code	2
5(b)(ii)	1 mark for drawback, 1 mark for appropriate matching expansion e.g. • Compatibility issues • ... the library routine may not work with current program • Not guaranteed to be thoroughly tested • ... there could be unexpected problems (bugs/virus) • The library routine may not match needs exactly // The library routine may give unexpected results • ... so it will need to be edited and tested • If the library routine is later edited or changed (by someone else) • ... there could be unexpected errors or results	2
5(c)	1 mark per bullet point, max 2 marks • Prettyprint • Expanding and collapsing code blocks • <u>Automatic</u> / <u>smart</u> indentation / formatting	2

7. O/N 25/P12/Q8/b(ii)

8(b)(ii)	1 mark per bullet point, max 2 marks • Disk repair / disk contents analysis • Back-up software	2
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8. O/N 25/P12/Q10/a

10(a)	1 mark per bullet point, max 4 marks Max 2 from each section. Interpreter: • Use an interpreter while writing/coding the program • ... to test / debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file so source code not seen • Use the compiler to repeatedly test the same completed section without having to re-interpret every time	4
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9. O/N 25/P13/Q8/a,b,d

8(a)	1 mark per bullet point, max 2 • Large amounts of source code take time to compile • It can be slower to produce the object code than an interpreter • The code must be recompiled when it is changed • The program cannot run if there are errors • It is not possible to correct errors in real-time • One error can cause false reporting of multiple further errors • Sections of code / unfinished code cannot easily be tested	2
8(b)	1 mark per bullet point, max 3 marks • Programming time is saved as code does not have to be written from scratch • Testing time is saved as code is already tested / documented • A library routine is more likely to work, as code is already tested • Library routines automatically update if they are changed / improved • The programmer can use library routines to perform complex functions / procedures that they may not be able to write themselves	3
8(d)	1 mark for each bullet point, max 3 marks • File management • Security management • Hardware / peripheral management // input / output management • Process management	3

10. O/N 25/P21/Q1/c

1(c)	One mark per point: - reference to a breakpoint - reference to single stepping - Correct explanation of both e.g. to stop program execution at specific point / line / statement / instruction and to execute one line / statement / instruction at a time to (locate an/the error) One mark for each bulleted point	3
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11. M/J 25/P11/Q6/b

6(b)	1 mark each to max 5 e.g. - To provide a user interface - so that the user is able to communicate with the hardware - To manage memory - so that data can be stored and accessed - and multitasking is possible - To manage files - allowing the user to create, edit, update and delete files and folders - To manage inputs and outputs from hardware/peripherals - To handle processes - to make sure each process has fair access	5
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12. M/J 25/P13/Q3/a

3(a)	1 mark for each correctly completed term an executable file/.exe source/program code stops immediately/in real-time A compiler checks all of the code before attempting to translate the program. If any errors are found, they are all reported at the same time and the program does not translate or run. If there are no errors found, the compiler produces an executable file/.exe which can run without access to the source/program code. An interpreter translates one line of code and then runs it, before moving to the next line of code. If the line of code has an error, the interpreter stops and displays the error. The programmer can correct the error immediately/in real-time and then the interpreter continues translating from that point.	4
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13. O/N 24/P11/Q4/a,c,d

4(a)(i)	1 mark for each bullet point (max 2) e.g. • Secondary storage space is divided into file allocation units • Space on secondary storage is allocated to particular files • Maintains/creates directory structures • Specifies the logical method of file storage e.g. FAT or NTFS • Provides file naming conventions • Controls user access // implements access rights //implements password protection // Makes file sharing possible • Controls access to the file by other software • Specifies tasks that can be performed on a file e.g. open, close, delete, copy, create, move, etc • Allows searching for a file	2
4(a)(ii)	1 mark for each bullet point (max 2) • To allow data to be retrieved / restored when lost • To automatically make a duplicate copy of data... • ...so the user does not have to remember to backup data • To make regular duplicate copies of data	2
4(c)	1 mark for each bullet point (max 3) e.g. • It saves development time for the student • ... by using pre-written code • There is less testing to do // the routines are more likely to work correctly • ... because they have already been tested and are error-free • It results in a more robust program • Subroutines can be used that they do not have the skill to write • ... which enables a more complex program to be written	3
4(d)(i)	1 mark for correct name, 1 mark for matching description • Expand/collapse code blocks... • ... sections of source code that are part of the same block can be expanded to see the content or collapsed so that the overall code is seen • Auto-indentation // auto-formatting... • ... automatically indents/formats code as the user types so that the structure is clear // aids readability	2
4(d)(ii)	1 mark for correct name, 1 mark for matching description • Breakpoints ... • ... stops the code running on a set line to view the current status/variable contents/program flow • Report window // variable watch window ... • ... shows the values in variables/data structures and how they change when each line is run	2

14. O/N 24/P12/Q3/c

3(c)	1 mark for each bullet point (max 4) e.g. • It manages the scheduling of processes // Decides which process is to be run next • Allows multi-tasking/multi-processing • Ensures fair access • Handles interrupts • Manages / allocates which resources the processes require • Facilitates the sharing and exchange of data between processes • Prevents interference between processes // conflict resolution	4
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15. O/N 24/P12/Q4/a,b

4(a)	1 mark for each bullet point (max 4) e.g. • Single stepping • Run the program one line at a time • ... and check the variable contents / program flow // show the effect of each line of code • Set breakpoints • ... run the code up to a set line • ... and then check the status • Variable/report watch window • ... view how the data changes as the program is running	4
4(b)	1 mark for each bullet point (max 3) e.g. • Subroutines can be shared / reused • ... between team members who are working independently • ... without having to rewrite/re-test them which saves the programmers' time • A program library provides continuity between programs/programmers • Individual programmers can contribute their specialisms to the library // Individual programmers can use the specialisms of others	3

16. O/N 24/P22/Q1/b,c

1(b)	For example: • Set a breakpoint at the start of / within <code>Lookup</code>, to stop execution at a given statement • then use single stepping to execute one statement / instruction at a time • to display the value of variables using a watch window One mark for each: MP1 Order starting with a breakpoint and an explanation – ‘stop execution at this statement / line’ MP2 Explanation of single stepping – execute ‘line by line’ / statements MP3 Explanation of watch window – displaying the value of <u>variable(s)</u>	3
1(c)	Features include: MP1 Editor MP2 <u>Auto</u>- (syntax) complete / <u>auto</u> correction // identify undeclared variables(s) MP3 Prettyprint / <u>auto</u>-indentation / <u>auto</u> (structure) highlighter MP4 <u>Dynamic</u> syntax checking MP5 Expand / collapse code blocks MP6 Context sensitive prompts Max 2 marks	2

17. M/J 24/P11/Q3/a,b

3(a)	1 mark each to max 2: • The interpreter will stop when an error is found • ... so the error can be corrected in real-time, and the result of changes seen immediately • Only one error is displayed at a time • ... so fewer errors to correct simultaneously and no dependent errors	2
3(b)	1 mark each to max 3: • Program can be distributed without source code • ... so it cannot be edited/stolen/plagiarised • Users do not require the translator to run the program • ... so time is not spent retranslating by user	3

18. M/J 24/P12/Q6

6	1 mark each to max 4 Max 3 marks for each management task: <i>Memory management: Max 3 marks</i> • Stores data from all currently running programs concurrently in RAM • Stops the data from overwriting each other in RAM/primary storage • Decides which processes should be in main memory • Makes efficient use of memory <i>Process management: Max 3 marks</i> • Allows one process to be paused whilst another process can be actioned • Decides which process is to be run next • Switches between processes to allow them to share the use of the processor • Identification/description of scheduling	4
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19. M/J 24/P12/Q8/a,b,c

8(a)	1 mark each to max 2: • Creates an executable file • ... so the code can be tested multiple times without having to recompile • ... so repeated testing takes less time	2
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8(b)	1 mark for identification of each feature and 1 mark for matching description: e.g. <i>For coding:</i> <table><tr><th>IDE feature</th><th>Description</th></tr><tr><td>Context-sensitive prompts</td><td>Gives suggestions for code as the user types instead of having to write/remember the code</td></tr><tr><td>Auto-correct</td><td>Corrects spelling mistakes so that user has fewer errors to correct</td></tr></table> <i>For presentation:</i> <table><tr><th>IDE feature</th><th>Description</th></tr><tr><td>Pretty-printing</td><td>Colour code keywords so the user can identify any errors</td></tr><tr><td>Expand/collapse (code) blocks</td><td>The user can hide code that they are not currently working on</td></tr></table> <i>For debugging:</i> <table><tr><th>IDE feature</th><th>Description</th></tr><tr><td>Single stepping</td><td>Run the code one line at a time // shows the effect of each line of code</td></tr><tr><td>Breakpoints</td><td>Stop the code running at a set point to check the flow/variable contents</td></tr></table>	IDE feature	Description	Context-sensitive prompts	Gives suggestions for code as the user types instead of having to write/remember the code	Auto-correct	Corrects spelling mistakes so that user has fewer errors to correct	IDE feature	Description	Pretty-printing	Colour code keywords so the user can identify any errors	Expand/collapse (code) blocks	The user can hide code that they are not currently working on	IDE feature	Description	Single stepping	Run the code one line at a time // shows the effect of each line of code	Breakpoints	Stop the code running at a set point to check the flow/variable contents	6
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Single stepping	Run the code one line at a time // shows the effect of each line of code																			
Breakpoints	Stop the code running at a set point to check the flow/variable contents																			
8(c)	1 mark each to max 2: • Saves programming/testing time as code does not have to be written/re-written from scratch // code does not have to be tested • Code is already tested so it is more robust/likely to work • The programmer does not need to maintain the library // library routines are updated automatically • Can perform complex calculations that the programmer may be unable to do • Makes code more easily readable	2																		

20. M/J 24/P13/Q7/e

7(e)(i)	1 mark for: Pre-written code/functions/routines that can be imported/called in another program	1
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7(e)(ii)	1 mark each to max 4 : • Maintenance not needed to be done by the programmer • ... because the DLL is separate from program • The calling program does not need recompilation by the programmer when a DLL file changes • ... because the DLL file can be updated independently of the calling program • ... updates will apply to all programs that use the DLL file	4
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21. O/N 23/P11/Q6/a,b,c

6(a)	1 mark for each bullet point (max 4) • easier to debug the program • ... because it translates line-by-line and stops when an error is found whereas the compiler translates all the program at the same time • ... only reporting one error at a time • ... which allows the error to be corrected in real time whereas the program would need to be corrected and recompiled • ... and the program can restart at same point when error occurred with a compiler the program needs to be re-run • The effect of any changes made by the programmer can be seen immediately with a compiler the effects can only be seen after re-running • A partially completed program can be translated / tested on its own a compiler cannot translate a partial program	4
6(b)	1 mark for each bullet point (max 1) • Partially compiled programs can be used on different platforms as they are interpreted when run • Code is optimised for the CPU as machine code is generated at run time • Source code does not need recompiling so more efficient to run	1
6(c)(i)	1 mark for each bullet point (max 2) • Prettyprint • Expand/collapse code blocks • Auto indentation / formatting	2
6(c)(ii)	1 mark for each bullet point (max 2) • Single stepping • Breakpoints • Report window • Variable expressions	2

22. O/N 23/P12/Q8/a,b

8(a)	1 mark for each bullet point (max 1) - To hide the complexities of the hardware from the user - To provide a platform for software to run - To provide a user interface	1
8(b)	1 mark for the name of the utility software 2 marks for the explanation - Defragmentation software ... because over time saving and deleting of small files fragments the disk ... the software makes (individual) files contiguous ... so access time to the files is improved ... because head movement is reduced - Disk contents analysis/disk repair software ... to identify and mark bad sectors ... to restore corrupted files ... to recover lost data (due to hardware failure) - File compression ... to reduce the size of files ... which saves storage and memory space // by example ... and reduces transmission time // by example - Disk formatter ... to prepare a disk for use // set up the file system ... to partition the disc ... to delete all the data from the disc	3

23. O/N 23/P13/Q5/a,b(ii)

5(a)	1 mark for each bullet point (max 2 for each feature). Context-sensitive prompts: - As the code is being written ...the options to complete the statement are shown Single stepping: - allows the programmer to execute the program one line at a time ...so that the effects of each statement can be seen	4
5(b)(ii)	1 mark for each bullet point (max 3). - Disk needs to be prepared for initial use - Disk needs to be checked for errors - A new file system needs to be generated on the disk - The file allocation table needs to be set up	3

24. M/J 23/P11/Q3/a,b,c

3(a)	1 mark each to max 4 Examples: • Installs device drivers • ... to allow communication between peripherals and computer • Sends data and receives data to and from peripherals • ... such as to an output device and from an input device/by example • Handles buffers for transfer of data • ... to ensure smooth transfer between devices that transmit and receive at different speeds • Manages interrupts / signals from the device	4
3(b)	1 mark each to max 2 • Memory management • File management • Security management • Process management • Error checking and recovery	2
3(c)	1 mark each to max 3 • Rearranges blocks of individual files (on the HDD) so they are contiguous // moves the free space together • Accessing each file is faster • ...because there is no need to search for the next fragment / block of the file • ...so less head movement is needed	3

25. M/J 23/P12/Q5/d

5(d)(i)	1 mark each to max 5 • Manages the scheduling of processes // decides which order to run processes • Manages which resources the processes require • ... such as allocating memory • Enables processes to share data • Prevents interference between processes // resolution of conflicts • Handles the process queue • It allows multi-tasking / multi-processing • ... by ensuring fair access, handling priorities and handling interrupts	5
5(d)(ii)	1 mark each to max 2 • To help users to set-up / configure / analyse / optimise / maintain the computer ... • ... by for example, making memory allocation more efficient • ... by for example, checking the system for faults	2

26. M/J 23/P12/Q7/a,b,c

7(a)(i)	1 mark each to max 2 • Set of pre-written / pre-compiled / pre-tested subroutines • ... which can be called in other programs • ... by installing/importing the library	2
7(a)(ii)	1 mark for each bullet point. Mark in pairs; 1 mark for a benefit and 1 mark for an appropriate expansion • (main) memory requirements for program is reduced • ... as dynamic link library is loaded only once / when required • the executable file size of the program using the DLL will be smaller • ... because the executable does not contain (all) the library routines • maintenance not needed to be done by the programmer • ... because the DLL is separate from program • no need to recompile the main program when changes are made to DLL • ... because changes / improvements/ error correction to the DLL file code are done independently of the main program • A single DLL file can be made available to several application programs • ... Saving space in memory / easing the pressure on memory	4
7(b)	No mark for choice. 1 mark each to max 3 for justification Interpreter • Allows the developer to make real-time changes • ... so the program can be debugged at each stage • ... the effect of any changes made by the developer can be seen immediately • The developer can test when incomplete • ... so small parts can be tested without having to test the rest of the program • ... if one section does not work others can still be tested • To avoid dependent errors Compiler • The developer can debug multiple errors simultaneously • Produces an executable file • ... so that the developer can test the program multiple times without recompiling	3

7(c)	1 mark each Breakpoints: • Stop the code at a specific line to check the current progress / values Dynamic syntax checks: • Highlight / underline / colour syntax errors as the code is entered Context-sensitive prompts: • Suggest the code to add // automatically complete statements Single stepping: • Run the code one line at a time so the values can be checked	4
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27. M/J 23/P13/Q5/a

5(a)(i)	1 mark each to max 2 • Programmer can test sections of the code without every part working / being written • Programmer can debug in real time • ... so that errors can be fixed and the program continued from that point • The effect of any changes made by the programmer can be seen immediately • To avoid dependent errors	2
5(a)(ii)	1 mark each to max 3 • The compiler produces an executable file • ... so the user cannot access / edit / sell the code • ... and users do not need the translator to run the game • The game can be compiled for different hardware specifications • ... and then used to generate more income for the programmer • The program can be tested multiple times without having to retranslate each time	3

28. M/J 23/P21/Q1/a

1(a)	For example: Could set a <u>breakpoint</u> to stop the program at a particular point / instruction then the value of variables could be checked using a <u>report/watch window</u> while <u>single stepping</u> can be used to execute one statement/line at a time. Marks available as follows: • One mark for each underlined term • One mark for an explanation of each term Note: max 4 marks	4
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29. M/J 23/P22/Q5/c

5(c)	Could use an IDE to: • Set a breakpoint to stop the program at a certain line / statement / point • Step through the program line by line / statement by statement • checking the value of 'num' / a variable using a report / watch window One mark per bullet	3
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30. O/N 22/P11/Q7/a,b

7(a)	1 mark for each benefit (max 2): • (main) memory requirements for program are reduced as dynamic link library is loaded only once / when required • the executable file size is smaller because the executable does not contain all the library routines • maintenance not needed to be done by the programmer because the DLL is separate from program • no need to recompile the main program when changes are made to DLL because changes / improvements/ error correction to the DLL file code are done independently of the main program	2
7(b)	1 mark for each bullet point (max 2): • RAM is assigned into blocks • dynamic allocation of RAM to programs / processes • reclaims unused blocks of RAM • prevents two programs / processes occupying the same area of RAM at the same time • moves data from secondary storage when needed // manages paging, segmentation and virtual memory	2

31. O/N 22/P12/Q1/a,b

1(a)	1 mark for each correct line. <table><thead><tr><th>Utility software</th><th>Purpose</th></tr></thead><tbody><tr><td>virus checker</td><td>to reorganise files so they are contiguous</td></tr><tr><td>disk formatter</td><td>to scan for malicious program code</td></tr><tr><td>backup</td><td>to decrease the file size</td></tr><tr><td>disk repair</td><td>to initialise a disk</td></tr><tr><td>defragmentation</td><td>to create copies of files in case the original is lost</td></tr><tr><td></td><td>to check for and fix inconsistencies on a disk</td></tr></tbody></table>	Utility software	Purpose	virus checker	to reorganise files so they are contiguous	disk formatter	to scan for malicious program code	backup	to decrease the file size	disk repair	to initialise a disk	defragmentation	to create copies of files in case the original is lost		to check for and fix inconsistencies on a disk	5
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1(b)(i)	1 mark for each bullet point (max 2): • larger amounts of source code take time to compile • slower to produce the object code than an interpreter • code cannot be changed without recompilation • the program will not run if there are any errors • errors cannot be corrected in real-time • one error may result in other false errors being reported • cannot easily test specific sections of the source code // cannot easily test unfinished source code	2														
1(b)(ii)	1 mark for each bullet point (max 2): • partially compiled programs can be used on different platforms as they are interpreted when run • code is optimised for the CPU as machine code is generated at run time	2														

32. O/N 22/P13/Q3

3	1 mark for each correct line: <table><tr><th>OS Management task</th><th>Description</th></tr><tr><td>hardware management</td><td>dynamically allocates memory to processes</td></tr><tr><td>security management</td><td>marks unallocated file storage for availability</td></tr><tr><td>memory management</td><td>installs programs for devices connected to external ports</td></tr><tr><td>process management</td><td>validates user and process authenticity</td></tr><tr><td></td><td>allows processes to transfer data to and from each other</td></tr></table>	OS Management task	Description	hardware management	dynamically allocates memory to processes	security management	marks unallocated file storage for availability	memory management	installs programs for devices connected to external ports	process management	validates user and process authenticity		allows processes to transfer data to and from each other	4
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	allows processes to transfer data to and from each other													

33. O/N 21/P11/Q4/b(ii),d

4(b)(ii)	1 mark for each correct tool e.g. • Colour coding // pretty printing • Auto-complete • Auto-correct • Context sensitive prompts • Expand and collapse code blocks	3
4(d)	1 mark for each correctly completed term Compilers are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program. They produce executable/.exe/object code files that can be run without the source code. Interpreters translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line. Assemblers are used to translate assembly code into binary/machine code.	4

34. O/N 21/P12/Q7/b,c

7(b)	1 mark for each bullet point to max 4 - Storage space is divided into file allocation units - Space is allocated to particular files - Maintains / creates directory structures - Specifies the logical method of file storage (e.g. FAT or NTFS) - Provides file naming conventions - Controls access // implements access rights // implements password protection // Makes file sharing possible - Specifies tasks that can be performed on a file (e.g. open, close, delete, copy, create, move etc.)	4
7(c)	1 mark for identifying program 1 mark for description, max 2 per program e.g. - Defragmentation - Less time is taken to access files because each one is contiguous so there is less head movement - Virus checker - makes more RAM available for programs to run ... because it removes software that might be taking up memory / replicating - Disk repair / Disk contents analysis - preventing bad sectors being used because it identifies / marks them - reduces access times by optimising storage - Disk/system clean up - releases storage by removing unwanted / temporary files	4

35. O/N 21/P13/Q4/b(ii),d

4(b)(ii)	1 mark for each correct tool e.g. - Colour coding // pretty printing - Auto-complete - Auto-correct - Context sensitive prompts - Expand and collapse code blocks	3
4(d)	1 mark for each correctly completed term Compilers are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program. They produce executable/.exe/object code files that can be run without the source code. Interpreters translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line. Assemblers are used to translate assembly code into binary/machine code.	4

36. M/J 21/P11/Q2/a,b

2(a)	1 mark for each correct line <table><thead><tr><th>Utility software</th><th>Description</th></tr></thead><tbody><tr><td>Disk formatter</td><td>Scans software for errors and repairs the problems</td></tr><tr><td>Defragmentation</td><td>Moves parts of files so that each file is contiguous in memory</td></tr><tr><td>Back-up</td><td>Creates a copy of data that is no longer required</td></tr><tr><td>Disk repair</td><td>Sets up a disk so it is ready to store files</td></tr><tr><td></td><td>Scans for errors in a disk and corrects them</td></tr><tr><td></td><td>Creates a copy of data in case the original is lost</td></tr></tbody></table>	Utility software	Description	Disk formatter	Scans software for errors and repairs the problems	Defragmentation	Moves parts of files so that each file is contiguous in memory	Back-up	Creates a copy of data that is no longer required	Disk repair	Sets up a disk so it is ready to store files		Scans for errors in a disk and corrects them		Creates a copy of data in case the original is lost	4
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	Scans for errors in a disk and corrects them															
	Creates a copy of data in case the original is lost															
2(b)	1 mark per bullet point to max 4 • memory management • file management • security management • hardware / device / peripheral / resources management • input/output management • process management • error checking and recovery • provision of a platform for software • provision of a user interface	4														

37. M/J 21/P12/Q7/a,b

7(a)	1 mark per bullet point to max 2 • Program libraries store pre-written functions and routines • The program library can be referenced/imported • the functions/routines can be called in her own program	2
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7(b)(i)	1 mark per bullet point to max 4; max 3 from each section Interpreter: • Use an interpreter while writing the program • ... to test/debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file • Use the compiler to repeatedly test the same (completed) section • ... without having to re-interpret every time // compiler not needed at run-time	4
7(b)(ii)	1 mark per correct tool to max 2 e.g. • Breakpoints • Single stepping • Report windows	2

38. M/J 21/P13/Q2/a,b

2(a)	1 mark for each correct line <table><tr><th>Utility software</th><th>Description</th></tr><tr><td>Disk formatter</td><td>Scans software for errors and repairs the problems</td></tr><tr><td>Defragmentation</td><td>Moves parts of files so that each file is contiguous in memory</td></tr><tr><td>Back-up</td><td>Creates a copy of data that is no longer required</td></tr><tr><td>Disk repair</td><td>Sets up a disk so it is ready to store files</td></tr><tr><td></td><td>Scans for errors in a disk and corrects them</td></tr><tr><td></td><td>Creates a copy of data in case the original is lost</td></tr></table>	Utility software	Description	Disk formatter	Scans software for errors and repairs the problems	Defragmentation	Moves parts of files so that each file is contiguous in memory	Back-up	Creates a copy of data that is no longer required	Disk repair	Sets up a disk so it is ready to store files		Scans for errors in a disk and corrects them		Creates a copy of data in case the original is lost	4
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2(b)	1 mark per bullet point to max 4 • memory management • file management • security management • hardware / device / peripheral / resources management • input/output management • process management • error checking and recovery • provision of a platform for software • provision of a user interface	4
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